SOLUTION
SECTION-I

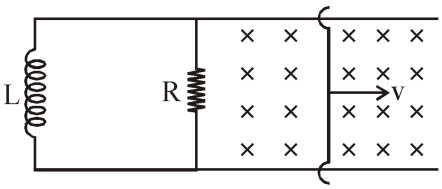
Single Correct Answer Type **8 Q. [3 M (–1)]**

1. Ans. (C)

Sol. $30 - L\frac{di}{dt} = i(20+5t) \quad \because \frac{di}{dt} = -ve \Rightarrow 30 - L\frac{di}{dt} > 30 \Rightarrow$ at t = 2sec ; i > 1 amp

2. Ans. (B)

Sol.



Work done by external goes in magnetic energy + heat dissipated.

$BLv = L\frac{di}{dt} \Rightarrow i = B\ell vt = Li$

$\therefore$ Energy in inductor $= \frac{1}{2}Li^2 = \frac{1}{2}L\frac{B^2v^2t^2\ell^2}{L^2} = 16\ J$

Energy dissipated in heat $= i^2Rt = \left(\frac{B^2\ell^2v^2}{R^2}\right)Rt = 16\ J$

3. Ans. (C)

Sol. Current through inductor remain same just after switch is opened

$\Rightarrow i = \frac{E}{r}$

Total heat generated $H = \frac{1}{2}Li^2 = \frac{1}{2}L\left(\frac{E}{r}\right)^2$

Heat generated in solenoid $= \frac{r}{r+R}[H] = \frac{E^2L}{2r(r+R)}$

4. Ans. (B)

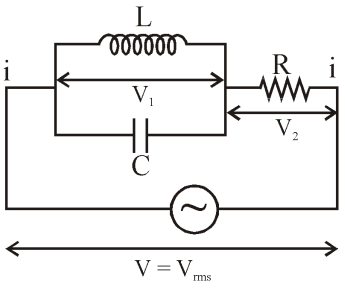
Sol. $V_1^2 + V_2^2 = V^2$

$\frac{V_1}{X_L} - \frac{V_1}{X_C} = i$

$V_1\left(\frac{1}{\omega L} - \omega C\right) = i$

$V_1 = \frac{\omega L\, i}{1-\omega^2 LC}$

$V_2 = iR$



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$$\frac{\omega^2L^2}{\left(1-\omega^2LC\right)^2}+R^2=\frac{V^2}{i^2}$$

$$i=\frac{V}{\sqrt{R^2+\frac{\omega^2L^2}{\left(1-\omega^2LC\right)^2}}}$$

5. Ans. (B)

Sol. $I=\sqrt{\left(\frac{V}{R}\right)^2+\left(\frac{V}{X_L}-\frac{V}{X_C}\right)^2}$

6. Ans. (D)

Sol. $\frac{f_{max}}{\Delta f_{\text{half of max power}}}=\text{Quality factor}$

$$\Rightarrow Q\approx\frac{7}{5}=1.4$$

7. Ans. (A)

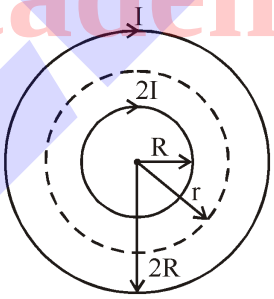
8. Ans. (B)

Sol. $I=Rt$

$B=\mu_0nRt$

$\phi=\pi R^2\times 2B+\pi r^2\times B$

$=(2R^2+r^2)\,\pi\mu_0nRt$



$$E(2\pi r)+\frac{d\phi}{dt}=\left(2R^2+r^2\right)\pi\mu_0nR$$

$$E=\left(\frac{2R^2+r^2}{r}\right)\frac{\mu_0nR}{2}$$

for charged particle

$$\frac{mv^2}{r}=qvB$$

$v=a_t\,t$

$$\boxed{v=\left(\frac{qE}{m}\right)t}$$

$$\frac{m}{r}\left[\frac{\left(2R^2+r^2\right)}{r}\frac{\mu_0nR}{2}\cdot\frac{q}{m}t\right]=q\mu_0nRt$$

$$\frac{2R^2+r^2}{2r^2}=1\qquad\left(r=\sqrt{2}\,R\right)$$

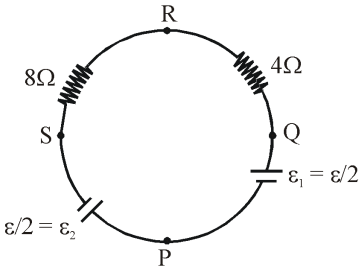
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Multiple Correct Answer Type **6 Q. [4 M (–1)]**

9. Ans. (A,C)

Sol.



Electrical equivalent :

$$i = \frac{\in_1 + \in_2}{4+8} \Rightarrow \frac{\in}{12} = 4$$

$$\in = 48 \text{ V}$$

$$\Delta V_{PR} = \in_1 - i(4) = 24 - 4(4) = 8V$$

10. Ans. (A,B,D)

Sol. $M = \frac{\phi_2}{i_1}, \phi_2 = BA$

where $B = \frac{\mu_0 i_1}{2\pi x}$

$A = 4 \times 10^{-6} \text{ m}^2$

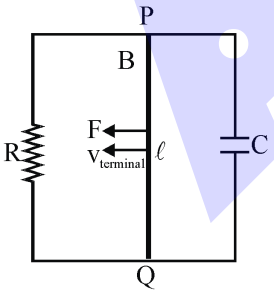
$$M = \frac{\mu_0}{2\pi x} \cdot A = \frac{2\times10^{-7}}{1} \times 4 \times 10^{-6} = 8 \times 10^{-13} \text{ H}$$

$i_1 = i_0 e^{-t/RC}$

$$e_2 = \frac{-d\phi_2}{dt} \& \; i_2 = \frac{e_2}{(8\times0.5)} \Rightarrow i_2 = \frac{\frac{M}{RC}\cdot\frac{q}{RC}e^{-t/RC}}{4} = \frac{8\times10^{-13}\times20\times10^{-6}}{(10\times10^{-6})^2\times4} e^{-\frac{2RC\ell n2}{RC}} = 10^{-8} \text{ A}$$

11. Ans. (BC)

Sol. In steady state



$$B\ell V_{terminal} = \frac{q_{max}}{C} \text{ also } iB\ell = F \; \& \; i = \frac{B\ell v_{terminal}}{R}$$

$$\Rightarrow \frac{B\ell v_T}{R} \cdot B\ell = F \Rightarrow v_T = \frac{FR}{B^2\ell^2} \text{ and } q_{max} = B\ell C \left(\frac{FR}{B^2\ell^2}\right)$$

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12. Ans. (B,D)

Sol. $i=\frac{E}{R}\left(1-e^{-Rt/L}\right)\Rightarrow\frac{di}{dt}=\frac{E}{L}e^{-Rt/L}$ = Slope in i-t graph

Circuit-1

$$\frac{di}{dt}=\frac{E_1}{L_1}e^0=\tan 30^\circ\Rightarrow\frac{E_1}{L_1}=\frac{1}{\sqrt{3}}$$

$$\text{Also } i_0=\frac{E_1}{R_1}$$

$$\Rightarrow\frac{i_0}{1/\sqrt{3}}=\frac{E_1/R_1}{E_1/L_1}\Rightarrow\frac{L_1}{R_1}=\sqrt{3}\,i_0=\tau_1$$

Circuit-2

$$\frac{di}{dt}=\frac{E_2}{L_2}e^0=\tan 60^\circ\Rightarrow\frac{E_2}{L_2}=\sqrt{3}$$

$$\text{Also } 2i_0=\frac{E_2}{R_2}$$

$$\Rightarrow\frac{2i_0}{\sqrt{3}}=\frac{E_2/R_2}{E_2/L_2}\Rightarrow\frac{L_2}{R_2}=\frac{2i_0}{\sqrt{3}}=\tau_2$$

$$\Rightarrow\frac{\tau_1}{\tau_2}=\frac{3}{2}$$

$$\text{Also } \left(\frac{di}{dt}\right)_2=3\left(\frac{di}{dt}\right)_1$$

13. Ans. (A,B,C)

Sol. $i_R=\frac{V}{R}$

$$\phi=\frac{\pi}{4}\Rightarrow i_L=i_R\Rightarrow\omega L=R$$

$$i=\sqrt{\left(\frac{V}{R}\right)^2+\left(\frac{V}{R}\right)^2}$$

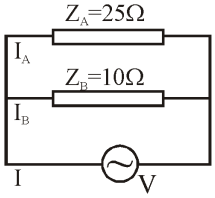
$$=\frac{\sqrt{2}100}{100}=\sqrt{2}A$$

14. Ans. (A,D)

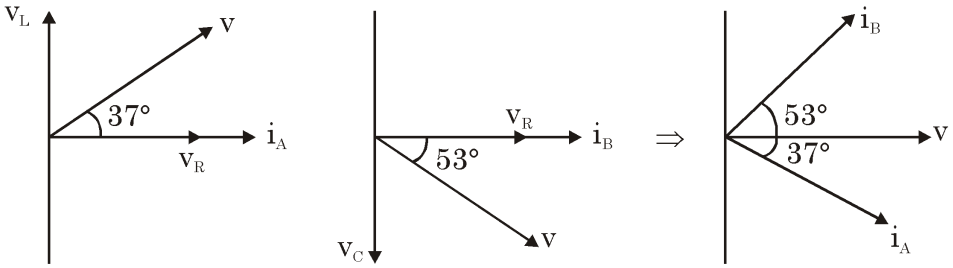
Linked Comprehension Type (2 Para × 2Q.) (2 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

15. Ans. (C)

Sol. Parallel circuit :



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Circuit A Circuit B

$$i = \sqrt{i_A^2 + i_B^2} \Rightarrow \left(\sqrt{29}\right)^2 = \left(\frac{v}{25}\right)^2 + \left(\frac{v}{10}\right)^2 \Rightarrow \frac{29v^2}{2500} = 29 \Rightarrow v = 50$$

$$i_A = \frac{50}{25} = 2A \text{ and } i_B = \frac{50}{10} = 5A$$

16. Ans. (B)

Sol. Power $= I_A^2 R_A + I_B^2 R_B = (2)^2(20) + (5)^2(6)$
$= 80 + 300 = 380$ W

17. Ans. (A)

Sol. EMF $= \pi b^2 \frac{dB}{dt} \Rightarrow K\pi b^2 = E(2\pi b)$

$$\Rightarrow F = qE = \frac{qKb}{2}$$

$$\tau = Fb = \frac{qKb^2}{2} = (mb^2)\alpha$$

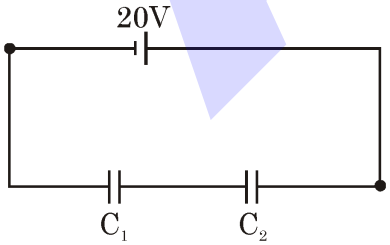
$$\Rightarrow \alpha = \frac{qK}{2m}$$

18. Ans. (B)

Sol. $v = (\alpha b)t = \frac{qKb}{2m}t$

19. Ans. (C)

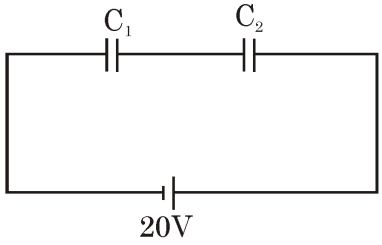
Sol. When k_1 closed & k_2 open



After very long time P.D
across both capacitor = 20 volt

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$$\frac{V_1}{V_2}=\frac{C_2}{C_1}=\frac{3}{2}$$

$$Q = CV \Rightarrow V \propto \frac{1}{C}$$

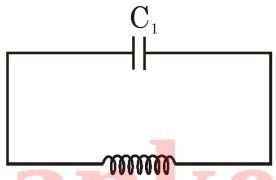
$$V_1 = \frac{3}{5}\times 20 = \frac{60}{5} = 12 \text{ volt}$$

$V_2 = 8$ volt

$Q = C_2V_2 = 3 \times 8 = 24\ \mu C$

20. Ans. (C)

Sol.



$Q = 24\ \mu C$

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2\times 2\times 10^{-6}}}$$

$$= \frac{1000}{2} = 500$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{500}$$

at t = 0, charge will be maximum across capacitor $\xrightarrow[\text{again max.}]{}$ $t = \frac{T}{2} \Rightarrow \frac{\pi}{500}$ sec

21. Ans. (A)

Sol. Maximum energy in inductor

$$\Rightarrow \frac{1}{2}Li^2 = \frac{Q^2}{2C} \Rightarrow \frac{\left(24\times 10^{-6}\right)^2}{2\times\left(2\times 10^{-6}\right)}$$

$$\frac{24\times 24\times 10^{-12}}{4\times 10^{-6}} = 0.144 \text{ mJ}$$

22. Ans. (C)

Sol. $$\Rightarrow Z = \sqrt{\left(\frac{1}{\omega C}\right)^2 + R^2}$$

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$$\Rightarrow i_{max} = \frac{v_{in}}{Z} \Rightarrow v_{out} = i \times R$$

$$v_{out} = \frac{v_{in}}{Z} \times R$$

$$\frac{v_{in}}{v_{out}} = \frac{Z}{R} = \frac{\sqrt{\left(\frac{1}{\omega C}\right)^2 + R^2}}{R}$$

23. Ans. (D)

Sol. $x_C = \frac{1}{\omega C} = x_C = \infty \Rightarrow$ all potential drop will be across capacitor

$$v_{out} = 0 \Rightarrow \frac{\Delta v_{out}}{\Delta v_{in}} = 0$$

24. Ans. (A)

Sol. $Z = \sqrt{\left(\frac{1}{\omega C}\right)^2 + R^2}$

$$i_{max} = \frac{v_{in}}{Z} \Rightarrow v_{out} = i x_C$$

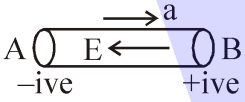
$$v_{out} = \frac{v_{in}}{Z} \times \left(\frac{1}{\omega C}\right)$$

$$\frac{v_{out}}{v_{in}} \frac{\frac{1}{\omega C}}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$$

Linked Comprehension Type (1 Para × 2 Q.) [4 M (–1)]
(Multiple Correct Answer Type)

25. Ans. (A, C)

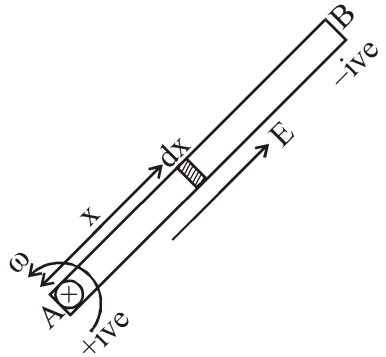
Sol. Due to pshedo force, electrons move from B to A.



$\Rightarrow v_B > v_A$

26. Ans. (A,C)

Sol. Due to centrifugal force electrons are moving from A to B.
At equilibrium, Electric force due to induce charges become equal to centrifugal force $m\omega^2 x = eE$



$$\int_0^\ell m\omega^2 x dx = \int_0^\ell eEdx = e\int_0^\ell Edx$$

$$\Rightarrow \frac{m\omega^2\ell^2}{2} = ev \quad (v = \text{voltage across rod})$$

$$\Rightarrow v = iR = \frac{m\omega^2\ell^2}{2e} \Rightarrow \frac{e}{m} = \frac{\omega^2\ell^2}{2iR}$$

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SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

27. Ans. 2

Sol. Magnetic field is non zero only in the region between the two solenoids , where $B = \mu_0 n_2 i_2$

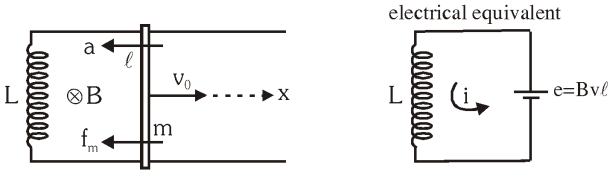
$\therefore$ energy stored per unit volume $= \dfrac{B^2}{2\mu_0} = \dfrac{\mu_0 n_2^{\,2} i_2^{\,2}}{2}$

The energy per unit length. = energy per unit volume × area of cross section were $B \neq 0$

$= \dfrac{\mu_0 n_2^{\,2} i_2^{\,2}}{2}\,[\pi\,(r_2^{\,2} - r_1^{\,2})] = \dfrac{\mu_0 n_1^2 i_1^2}{2}\,[\pi(r_2^{\,2} - r_1^{\,2})$, since $n_1 i_1 = n_2 i_2$

28. Ans. 5

Sol.



According to Len's law, magnetic force, will oppose the motion of conducting rod.

$F_m = i\ell B$; $ma = i\ell B$ and $\left(a = -\dfrac{dv}{dt}\right)$ so, $m = \left(-\dfrac{L}{B\ell}\dfrac{d^2i}{dt^2}\right) = i\ell B$ or $\dfrac{d^2i}{dt^2} = -\left(\dfrac{B^2\ell^2}{mL}\right)i$

and $\varepsilon = L\dfrac{di}{dt}$; $Bv\ell = L\dfrac{di}{dt}$; $(B\ell)\left(\dfrac{dv}{dt}\right) = L\dfrac{d^2i}{dt^2}$

This is a second order differential equation; So, $i = i_0 \sin(\omega t)$

where $\dfrac{1}{2}mv_0^2 = \dfrac{1}{2}Li_0^2 \Rightarrow \left[i_0 = v_0\sqrt{\dfrac{m}{L}}\right]$ and $\left[\omega = \sqrt{\dfrac{B^2\ell^2}{mL}}\right]$

We have, $Bv\ell = L\dfrac{di}{dt}$ or B (vdt) ℓ = L(di) or Bℓ (dx) = L(di)

or $x = \dfrac{Li}{B\ell} = \dfrac{Li_0 \sin\omega t}{B\ell} = \dfrac{Lv_0\sqrt{\dfrac{m}{L}}}{B\ell}\sin\omega t \Rightarrow x = \dfrac{v_0}{\omega}\sin\omega t$

So amplitude $= \dfrac{v_0}{\omega} = \dfrac{v_0}{B\ell}\sqrt{mL} = 5cm$

29. Ans. 5

Sol. $V = \dfrac{Br^2\omega}{2}$; $I = \dfrac{V}{R} = \dfrac{Br^2\omega}{2R}$; $\dfrac{B^2r^3\omega}{2R}\cdot\dfrac{r}{2} - mg\rho = 0$; $\omega = \dfrac{4mg\rho R}{B^2r^4} = 100s^{-1}$

Subjective Type **1 Q. [4 M (0)]**

30. Ans. $\dfrac{k\,h}{4\rho}(b^2 - a^2)$

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